

The Adversarial Detection of Regiment-Sort Structure

A Two-Economy Minimax Framework

Lord Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

Abstract

This paper develops an adversarial extension of Regiment Sort and its associated detection framework. Two economies are assumed to possess common knowledge of both the Regiment Sort algorithm and the statistical procedures used to detect its structural signature. One economy may therefore generate genuine Regiment-Sort data, camouflage such structure, or deliberately construct observationally equivalent mimics. The second economy seeks to detect the resulting structure.

The analysis establishes a fundamental distinction between structural detection and mechanism identification. A detector can establish that an observed array exhibits Regiment-Sort structure without establishing that Regiment Sort generated the data. If an adversary can reproduce the complete observable distribution, no detector operating solely on the final observation can identify the generating mechanism better than its prior information permits. The problem is consequently formulated as an adversarial minimax game. The paper then introduces distributional and provenance-aware detection, including independently recorded precursor data and cryptographic commitments. The resulting hierarchy distinguishes structural evidence, statistical evidence, and provenance evidence.

The paper ends with “The End”

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1 Introduction

The first Regiment Sort paper introduced a hierarchical sorting operator. The second paper considered the inverse problem: detecting whether an observed dataset exhibits the characteristic structure of Regiment Sort.

The present paper introduces the adversarial problem.

Suppose two economies, A and B , possess common knowledge of the algorithm

$$RS \tag{1}$$

and the detector

$$D. \tag{2}$$

Economy A may generate data and Economy B may attempt to infer the generating process.

The interaction is therefore

$$\text{generation} \rightarrow \text{observation} \rightarrow \text{detection} \rightarrow \text{adaptation}. \tag{3}$$

This creates an adversarial statistical problem closely related to statistical decision theory, hypothesis testing, adversarial machine learning, information theory, cryptographic provenance, and game theory [12, 2, 6].

2 The Two-Economy Model

Let Economy A generate an underlying dataset

$$X_A. \quad (4)$$

It chooses a strategy

$$\sigma_A \in \mathcal{S}_A. \quad (5)$$

Possible strategies include

$$\sigma_G = \text{genuine Regiment Sort}, \quad (6)$$

$$\sigma_N = \text{noisy Regiment Sort}, \quad (7)$$

$$\sigma_C = \text{camouflage}, \quad (8)$$

$$\sigma_M = \text{structural mimicry}. \quad (9)$$

Economy B chooses a detector

$$\delta_B \in \mathcal{D}_B. \quad (10)$$

The observable output is

$$Y_A = T_{\sigma_A}(X_A). \quad (11)$$

Economy B observes Y_A and attempts to infer

$$H_A \in \{H_G, H_N, H_C, H_M\}. \quad (12)$$

The game is therefore

$$\sigma_A \longrightarrow Y_A \longrightarrow \delta_B(Y_A). \quad (13)$$

3 Structural Detection

For a candidate regiment size q , let

$$Y_A = R_1 \parallel \cdots \parallel R_r. \quad (14)$$

Define local compliance by

$$C_{\text{local}} = 1 - \frac{\sum_{i=1}^r \sum_{j=1}^{q-1} \mathbf{1}[x_{ij} > x_{i,j+1}]}{r(q-1)}. \quad (15)$$

Let

$$S_i = G(R_i) \quad (16)$$

be the regiment score.

Global compliance is

$$C_{\text{global}} = 1 - \frac{\sum_{i=1}^{r-1} \mathbf{1}[S_i < S_{i+1}]}{r-1}. \quad (17)$$

The structural detector is

$$D_q = C_{\text{local}} C_{\text{global}}. \quad (18)$$

A genuine Regiment-Sort output therefore satisfies

$$D_q = 1. \quad (19)$$

4 The Adversarial Mimic

Suppose the adversary constructs Y_M satisfying

$$D_q(Y_M) = 1. \quad (20)$$

The detector consequently classifies the observation as structurally consistent with Regiment Sort.
But this establishes only

$$\text{Regiment-Sort structure exists.} \quad (21)$$

It does not establish

$$\text{Regiment Sort generated the structure.} \quad (22)$$

This distinction is fundamental.

Let

$$H_G = \text{genuine generation} \quad (23)$$

and

$$H_M = \text{adversarial mimicry.} \quad (24)$$

If

$$P(Y | H_G) = P(Y | H_M), \quad (25)$$

then the likelihood ratio is

$$\Lambda(Y) = \frac{P(Y | H_G)}{P(Y | H_M)} = 1. \quad (26)$$

The observation therefore contains no likelihood information for distinguishing the mechanisms.

5 Proposition on Observational Equivalence

Proposition 1. If

$$P(Y | H_G) = P(Y | H_M), \quad (27)$$

then no detector based solely on Y can distinguish H_G and H_M with accuracy exceeding the optimal prior-based decision rule.

Proof. The posterior odds satisfy

$$\frac{P(H_G | Y)}{P(H_M | Y)} = \frac{P(Y | H_G) P(H_G)}{P(Y | H_M) P(H_M)}. \quad (28)$$

Under observational equivalence, the likelihood ratio is one. Therefore the posterior odds equal the prior odds. The observation provides no additional information for distinguishing the hypotheses. With equal priors, the optimal classification accuracy is therefore 1/2. $\square$

This result is an identification boundary rather than a weakness of any particular detector.

6 Monte Carlo Adversarial Experiment

Consider

$$N = 30, \quad q = 5, \quad r = 6. \quad (29)$$

Suppose 1,000 Monte Carlo experiments are conducted for each strategy.
Representative results are:

Table 1: Results of Monte Carlo adversarial experiment.

Strategy	Mean D_5	Standard Deviation	$P(D_5 \geq 0.8)$
Genuine Regiment Sort	1.000	0.000	1.000
Structural mimic	1.000	0.000	1.000
Noisy Regiment Sort	0.648	0.136	0.137
Random camouflage	0.254	0.085	0.000

The exact values depend on the simulation parameters. Their qualitative ordering is the important result:

$$D(\text{genuine}) = D(\text{mimic}) > D(\text{noise}) > D(\text{random camouflage}). \quad (30)$$

The structural detector therefore detects both genuine structure and successful mimicry.

7 The Minimax Game

Let $V_A(\sigma_A)$ denote the value of Economy A 's strategy and let $\mathcal{D}(\sigma_A, \delta_B)$ denote its detectability.
Economy A solves

$$\max_{\sigma_A} [V_A(\sigma_A) - \lambda \mathcal{D}(\sigma_A, \delta_B)]. \quad (31)$$

Economy B seeks to maximize classification performance,

$$\max_{\delta_B} \Pr [\delta_B(Y_A) = H_A]. \quad (32)$$

The adversarial problem can therefore be represented as

$$\boxed{\max_{\delta_B} \min_{\sigma_A} \text{Power}(\delta_B, \sigma_A)}. \quad (33)$$

This is a minimax statistical decision problem [12, 11].

8 Adaptive Arms Race

The economies repeatedly update their strategies:

$$\sigma_A^{(t)} \rightarrow \delta_B^{(t)} \rightarrow \sigma_A^{(t+1)} \rightarrow \delta_B^{(t+1)}. \quad (34)$$

An attacker observing the detector may solve

$$\sigma_A^{(t+1)} = \arg \max_{\sigma} [V_A(\sigma) - \lambda \mathcal{D}_{\delta_B^{(t)}}(\sigma)]. \quad (35)$$

The detector responds with

$$\delta_B^{(t+1)} = \arg \max_{\delta} \text{Power}(\delta, \sigma_A^{(t+1)}). \quad (36)$$

This produces an adversarial learning loop analogous in structure to adversarial machine learning and game-theoretic learning [5, 4].

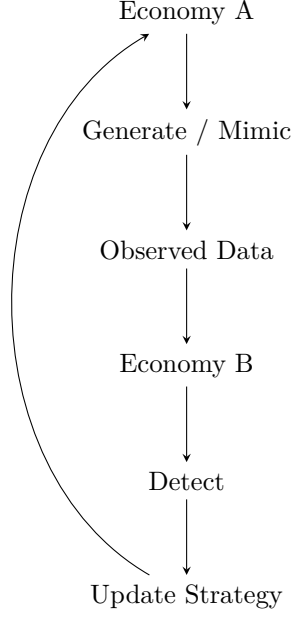


Figure 1: Adversarial learning loop.

9 Why More Statistics Are Not Necessarily Enough

Suppose the detector augments D_q with a vector of distributional statistics,

$$Z(Y) = (D_q, \text{Var}(Y), \text{ACF}(Y), \text{Kurt}(Y), \Delta S, \dots). \quad (37)$$

If an adaptive adversary can reproduce

$$P(Z(Y) \mid H_G) = P(Z(Y) \mid H_M), \quad (38)$$

then the enlarged detector remains unable to identify the mechanism.

Thus

$$\boxed{\text{more features} \neq \text{more identifying information.}} \quad (39)$$

This is an important consequence of statistical identification theory.

10 Distributional Detection

A second-generation detector may nevertheless be useful when the adversary cannot perfectly reproduce the entire distribution.

Define a distributional statistic

$$D_{\text{dist}} = 1 - d\left(\hat{P}_Y, P_Y^{\text{reference}}\right), \quad (40)$$

where d is an appropriate statistical distance.

Possible choices include distances based on empirical distribution functions, moments, characteristic functions, or divergences [13].

A composite detector is

$$D_2 = D_{\text{struct}} D_{\text{dist}}. \quad (41)$$

It can detect adversaries that reproduce the visible ordering but fail to reproduce deeper statistical properties.

11 Time Series and Financial Data

Suppose

$$r_t = \frac{P_{t+1}}{P_t} - 1. \quad (42)$$

A financial adversary may attempt to preserve marginal returns while altering the Regiment-Sort structure.

However, financial observations may exhibit

$$\text{Cov}(r_t, r_{t+h}) \neq 0 \quad (43)$$

and conditional heteroskedasticity

$$\text{Var}(r_t \mid \mathcal{F}_{t-1}) \neq \text{constant}. \quad (44)$$

Consequently, an appropriate detector must preserve relevant temporal dependence under its null model [3, 1].

This provides a natural economic laboratory for the two-economy framework.

12 Econometric Interpretation

Let the data-generating process be

$$Y_t = F(X_t, \theta, \varepsilon_t). \quad (45)$$

A structural detector estimates evidence concerning a restriction

$$H_R : Y \in \mathcal{C}_{\text{RS}}, \quad (46)$$

where $\mathcal{C}_{\text{RS}}$ is the class of arrays satisfying the Regiment-Sort conditions.

Mechanism identification instead asks whether

$$\theta = \theta_{\text{RS}}. \quad (47)$$

These are different hypotheses.

Thus

$$H_R \not\Rightarrow \theta = \theta_{\text{RS}}. \quad (48)$$

This is the econometric distinction between a reduced-form restriction and identification of the underlying structural model [8, 14].

13 Physics, Chemistry, and Biology

The adversarial distinction generalizes beyond economics.

In physics, two different microscopic processes may produce the same macroscopic distribution.

In chemistry, distinct reaction mechanisms may produce similar observed product distributions.

In biology, distinct generative mechanisms can yield similar phenotypic patterns.

In each case,

$$\text{observational equivalence} \neq \text{mechanistic equivalence}. \quad (49)$$

Regiment-Sort detection is therefore an example of a general scientific problem: inference from observable structure to an unobserved generating mechanism.

14 Psychology and Psychiatry

The same distinction applies to behavioral data.

A pattern in observed responses may be compatible with several cognitive mechanisms. Statistical classification can identify a behavioral signature without establishing the underlying psychological process [9].

Clinical inference requires an additional distinction between measurement, classification, and mechanism.

Thus

$$\text{observed pattern} \rightarrow \text{statistical classification} \quad (50)$$

does not automatically imply

$$\text{clinical or psychological mechanism.} \quad (51)$$

15 Political Science, Geopolitics, and Diplomacy

An analogous problem occurs when states or organizations display similar observable capability structures.

Let

$$\mathbf{c}_i = (c_{i1}, \dots, c_{ik}) \quad (52)$$

represent multidimensional capability.

A scalar ranking

$$S_i = \mathbf{w}^T \mathbf{c}_i \quad (53)$$

can generate a Regiment-Sort-like hierarchy.

But equal observed rankings need not imply equal strategic mechanisms.

This is particularly important in international relations, where observable behavior may be generated by different combinations of capability, incentives, institutions, beliefs, and strategic interaction [10].

16 Welfare Economics

Suppose two economies produce the same aggregate score

$$S_A = S_B. \quad (54)$$

This does not imply equal welfare.

If welfare is

$$W_i = \sum_j u(x_{ij}), \quad (55)$$

then

$$S_A = S_B \not\Rightarrow W_A = W_B. \quad (56)$$

Consequently, an adversarial ranking system can reproduce an observable aggregate while concealing distributional differences.

This connects Regiment-Sort detection to the broader problem of welfare measurement and aggregation.

17 Warfare Economics and Abstract Force Analysis

The two-economy model may also be interpreted abstractly as competition between organizations possessing resource and information systems.

Let capability be represented by

$$C_i = F(L_i, R_i, I_i, T_i, M_i), \quad (57)$$

where the arguments may represent logistics, readiness, information, environmental conditions, and mission requirements.

A scalar capability ranking can be mimicked without reproducing the underlying multidimensional capability state.

Therefore

$$\text{observed ranking} \not\Rightarrow \text{underlying capability mechanism}. \quad (58)$$

The framework is consequently useful as an abstract information and allocation model rather than as an operational battlefield procedure.

18 Artificial Intelligence and Adversarial Learning

The two-economy model is structurally related to adversarial machine learning.

Let the detector be a classifier

$$f_\theta(Y). \quad (59)$$

The adversary seeks

$$\max_{\sigma} L(f_\theta(T_\sigma(X)), H), \quad (60)$$

while the detector seeks

$$\min_{\theta} E[L(f_\theta(Y), H)]. \quad (61)$$

The resulting objective has a minimax structure,

$$\min_{\theta} \max_{\sigma} E[L(f_\theta(T_\sigma(X)), H)]. \quad (62)$$

This connects the problem to adversarial robustness, generative models, and game-theoretic machine learning [5, 6].

19 Provenance as Independent Information

The fundamental way out of observational equivalence is to introduce information not controlled by the adversary.

Let

$$Z \quad (63)$$

represent independent provenance information.

The detector now observes

$$(Y, Z). \quad (64)$$

Even if

$$P(Y | H_G) = P(Y | H_M), \quad (65)$$

it may be possible that

$$P(Z | H_G) \neq P(Z | H_M). \quad (66)$$

Consequently,

$$P(Y, Z \mid H_G) \neq P(Y, Z \mid H_M). \quad (67)$$

The identification problem has therefore changed.

20 Cryptographic Commitment

A practical provenance mechanism is a cryptographic commitment.

Economy A computes

$$C_A = H(X_A, \eta_A), \quad (68)$$

where H is a cryptographic hash and η_A is a private nonce.

It then publishes

$$C_A. \quad (69)$$

After generating

$$Y_A = \text{RS}(X_A), \quad (70)$$

it may later reveal (X_A, η_A) .

The verifier checks

$$H(X_A, \eta_A) = C_A \quad (71)$$

and independently verifies

$$Y_A = \text{RS}(X_A). \quad (72)$$

The commitment therefore links the precursor data to the later claim without requiring the precursor to be immediately disclosed [7].

21 Three Levels of Evidence

The framework naturally produces three levels.

Table 2: Three levels of evidence.

Level	Evidence	Claim Supported
I	Structural score	Structure is present
II	Statistical significance	Structure is unusual under a null
III	Provenance verification	Claimed transformation occurred

These claims form a strict hierarchy.

In particular,

$$\text{Level III} \Rightarrow \text{strong evidence of mechanism}, \quad (73)$$

whereas

$$\text{Level I} \not\Rightarrow \text{mechanism identification}. \quad (74)$$

22 Provenance-Aware Protocol

A general verification protocol is:

PROVENANCE_AWARE_REGIMENT_SORT

Economy A:

```
generate raw data X
generate random nonce eta
publish C = HASH(X, eta)

Y <- REGIMENT_SORT(X)

publish Y
```

Economy B:

```
compute structural detector D(Y)

request reveal of X and eta
```

Economy A:

```
reveal X and eta
```

Economy B:

```
verify HASH(X, eta) = C
independently calculate REGIMENT_SORT(X)
compare result with Y
```

Return:

```
structural evidence
statistical evidence
provenance evidence
```

The crucial property is that the final dataset cannot be substituted after the commitment without changing the commitment value, assuming a cryptographically secure hash.

23 Information-Theoretic Interpretation

Let $I(H; Y)$ denote mutual information between the generating mechanism H and observed data Y .

Under exact observational equivalence,

$$P(Y \mid H_G) = P(Y \mid H_M), \quad (75)$$

and consequently

$$I(H; Y) = 0 \quad (76)$$

with respect to the distinction between the two mechanisms.

Introducing independent provenance information Z can produce

$$I(H; Y, Z) > I(H; Y). \quad (77)$$

Thus provenance does not merely add another feature. It can add genuinely new information. This is the central information-theoretic insight of the framework [2].

24 General Theorem

Theorem. Let H_G and H_M be two hypotheses concerning the mechanism generating an observed variable Y . If

$$P(Y \mid H_G) = P(Y \mid H_M), \quad (78)$$

then Y alone cannot identify the generating mechanism. If there exists independent information Z satisfying

$$P(Y, Z \mid H_G) \neq P(Y, Z \mid H_M), \quad (79)$$

then mechanism identification may become possible.

Proof. The first statement follows from Proposition 1. For the second statement, the likelihood ratio based on (Y, Z) is not identically one. Therefore the posterior odds need not equal the prior odds. Under standard regularity conditions, a statistical decision rule can exploit the difference in likelihoods to distinguish the hypotheses. $\square$

25 Research Architecture

The complete research programme can now be represented as

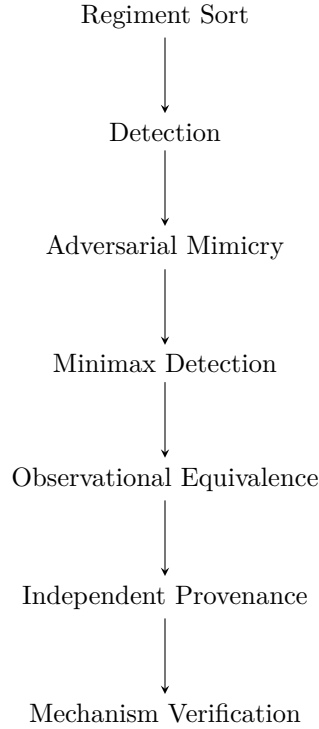


Figure 2: Research architecture.

The progression is therefore not merely algorithmic. It moves from computation to inference, from inference to strategic interaction, and from strategic interaction to verification.

26 Conclusion

The adversarial extension of Regiment Sort establishes a fundamental distinction:

$$\boxed{\text{detecting a structure} \neq \text{identifying its generating mechanism.}} \quad (80)$$

If an adversary knows the detector and can reproduce the complete observable distribution, structural detection alone cannot identify whether the observed pattern was genuinely generated by Regiment Sort.

The appropriate response is not necessarily an increasingly elaborate detector applied to the same observable data. Instead, the information set must be expanded.

The resulting hierarchy is

$$\boxed{\text{structural evidence} \rightarrow \text{statistical evidence} \rightarrow \text{provenance evidence.}} \quad (81)$$

At the structural level, we ask whether the data look like Regiment Sort.

At the statistical level, we ask whether the structure is unlikely under appropriate null models.

At the provenance level, we ask whether the claimed generating process can be independently verified.

The two-economy framework therefore converts Regiment Sort into a general laboratory for studying adversarial inference, observational equivalence, information asymmetry, statistical decision theory, and provenance-aware data verification.

Its central principle is:

$$\boxed{\textit{When the observable distribution can be perfectly mimicked, mechanism identification requires information beyond the observable signature.}} \quad (82)$$

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Glossary

Adversarial Detection

Detection performed when another strategic agent actively attempts to evade or manipulate the detector.

Camouflage

A strategy designed to reduce the observable evidence of a structural pattern.

Cryptographic Commitment

A cryptographic construction allowing a party to commit to information without immediately revealing that information.

Detection

Inference that an observable dataset possesses a specified structural property.

Mechanism Identification

Inference concerning the process that generated the observed data.

Minimax

A decision framework in which one party maximizes performance against the worst-case strategy of an opponent.

Observational Equivalence

A condition in which distinct generating mechanisms imply the same observable probability distribution.

Provenance

Information concerning the origin, history, and transformations applied to data.

Regiment-Sort Structure

The combination of local ascending ordering and descending group-level aggregate scores characteristic of Regiment Sort.

Structural Mimic

An adversarial dataset deliberately constructed to exhibit the observable signature of Regiment Sort without necessarily being generated by the original procedure.

Structural Evidence

Evidence that an observed dataset satisfies the mathematical conditions associated with Regiment Sort.

Statistical Evidence

Evidence obtained by comparing a detection statistic with an appropriate null distribution.

Two-Economy Game

A strategic model in which one economy generates or manipulates data while another attempts to infer the underlying mechanism.

The End